

EVALUATING PYTORCH PERFORMANCE IN DATA-CENTRIC AI: A COMPARATIVE ANALYSIS OF DYNAMIC GRAPHS AND COMPUTATIONAL THROUGHPUT

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ABSTRACT

This paper explores the role of the PyTorch framework in the Data-Centric AI field where it focuses on data quality and data handling factors. The main objective of this paper is to evaluate how the PyTorch framework, which is dynamic computational graphs helps to improve overall system performance and flexibility compared to legacy systems like TensorFlow. The research uses a qualitative method by reviewing the past related academic papers and analyzing a real-world case study about wildflower monitoring. This paper also shows that PyTorch allows for faster error fixing and better data cleaning as it is like normal Python programming and able to build its math map while operating. The study concludes that PyTorch is a forward technology for modern data research as it provides high throughput and user-friendly designation. However, the study also states that the system may use more computer memory which should take further development of technology like TorchDynamo into consideration.

Keywords: Data-Centric AI, PyTorch, dynamic, data processing

1.0 INTRODUCTION

Huge amounts of data are being generated very fast by big organizations day by day. Millions of pieces of information are being collected from sources like social media, smart devices and sensors. According to Jakubik et al (2024), experts focus more on Data-Centric AI which means instead of trying to adjust a better math model, they shift their focus on making sure the data that will be used is high quality and correctly handled data. However, there are huge issues when the individual tries to process the data.

Traditional technology that is used for AI were often built with a static structure which means the system is very strict and inflexible. These traditional tools cannot match with the real-time business analytics for instance, as they operate at a slower speed and do not meet the expected processing speeds (Kumar et al., 2024). Therefore, there is a clear gap between how fast the data is growing in today's world and how fast the old technology can process the data.

Data-Centric AI systems are very important as it reflects how fast we can process the data using the useful technology that exists from time to time. Advanced technologies enable the experts to receive better results more efficiently while researchers are able to execute their hands-on experiments without extended waiting times. By using the right technology, it helps the organizations to ensure the data is huge in amount and useful for decision making (Dagino, 2024).

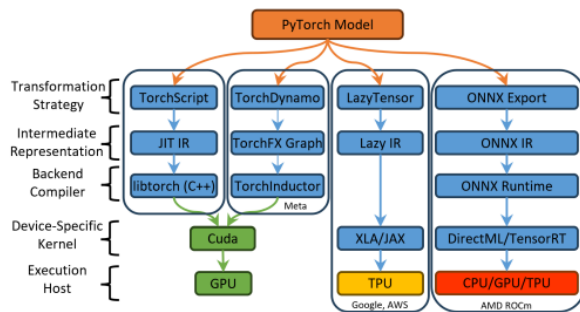
This paper aims to analyze the PyTorch technology framework and examine its performance in Data-Centric AI systems. This paper will specifically focus on how the PyTorch uses dynamic graphs to help the data move through the system at high speed. The paper is organized into several parts, the first part will explain the PyTorch technology and its design, second on the main body that includes a real-world case study, and last will be a comparative analysis study of the technology.

2.0 TECHNOLOGY DESCRIPTION AND ARCHITECTURE

PyTorch is a popular technology that is commonly used for high performance data processing as its simple functionality to write and read, almost similar to Python code (Maczynski & Ganguly, 2024). As in Figure 1, the PyTorch architecture is organized into a multi-layered system which

transforms high-level code into hardware prepared instructions. The framework uses multiple transformation strategies to capture the model logic.

Figure 1. Back-end Compilation and Execution Stack of PyTorch.



A key feature of the architecture is the TorchDynamo which is a bytecode tracer that interrupts the execution process to create a TorchFX Graph. This approach is also known as the “Define-by-Run” as it allows the system to be very adjustable to the graph whenever there are changes of data (Li et al., 2024). Besides, the stack in the Figure 1 shows how the models are decreased to Device-Specific Kernels like CUDA. It allows the system to work directly with GPUs to increase the computational throughput and allows it to process huge datasets in a short time.

3.0 IMPLEMENTATION OF PYTORCH FOR AUTOMATED ECOLOGICAL MONITORING

A good sample of PyTorch case study for real data work is a project that monitors the wildflowers in nature ecosystems. The researchers have found out different types of flowers across various fields using photos taken from up skies (Schouten et al., 2024). It involves massive amounts of data as the photos are large and need to be processed for details. The researchers used PyTorch technology to process the images and identify each species of flower automatically. According to the research, the implementation was successful because the PyTorch allows them to handle data in a flexible approach. Since the images were very large in size, it is harder for computer to process all at once without crashing. Due to this issue, they used a method called “tiling”, where the large size of image is cut into small pieces so that PyTorch is able to process these small pieces one by one without losing its quality. This shows that the PyTorch framework able to manage difficult data workloads more effectively.

In addition, study showed that having good data is just as important as having a complex model. By using PyTorch, the researchers are able to clean their data and fix errors very quickly. This approach helps to ease their works and provide accurate results by focusing solely on the images. The speed of PyTorch technology was also very helpful for this ecological project as there were tons of photos to crosscheck and they needed a system that could move the data they gain through the processor in a faster way. High throughput of this PyTorch technology meant that the researchers did not have to wait for days to see their results and instead, they could get the numbers of flower types quickly which saved a lot of their time.

4.0 COMPARATIVE ANALYSIS BETWEEN STATIC EXECUTION AND DYNAMIC GRAPH FLEXIBILITY

This section will show a better comparison of why PyTorch is a great choice of technology compared to other traditional systems that use static graphs. In the legacy system, a whole data plan needs to be built before starting the program. If the data keep changing or has a small error, the whole legacy system might stop functioning and it would be hard to solve it. Therefore, to solve this, PyTorch is a better solution by being “dynamic”. It functions to be more flexible as you can either skip the errors of the data or solve it to pass through it. This flexibility is a key aspect in building a modern AI as the data comes in messy forms and changes all of the time. By having this function which is able to change the math map while the program operates, it helps to save a lot of time for engineers (Li et al., 2024).

Another aspect to focus on is how easy it is for humans to use these technologies. Older systems are often designed to be more complex and hard to understand, meanwhile PyTorch is designed to be easier for beginners as it uses familiar programming language which is Python. Besides, the way that systems handle computer memory is also different as the old static way keeps repeating the same task and is not good at trying a new thing, meanwhile PyTorch built for research and its innovation. The speed of the technology and ease of making changes make it much better for modern data-centric where data cleaning and testing need to be done many times (Dagino, 2024).

Table 1. Technical Comparison of AI Frameworks between Traditional Static Systems and Modern Dynamic Systems.

Comparison Feature	Traditional Static Systems	Modern Dynamic System
Logic Style	Define-then-Run	Define-by-Run
Fixing Errors	Hard as errors are hidden	Easy, works like a normal code
Data Quality	Hard to adjust bad data	Easy to clean and test data
Ease of Use	Complex and strict	Simple and human-friendly
Research Speed	Slower due to strict design	Faster for testing new ideas

The key difference that is shown in Table 1 is the Logic Style as PyTorch uses Define-by-Run which means it is able to adapt the cleaned and updated data immediately. Meanwhile the older systems are fixed as the systems might crash if the data keeps changing which is not time efficient. For the Fixing Errors, PyTorch is much better as it shows the developer the exact problem as soon it counters, meanwhile for the old systems, the error might stay hidden inside the complex static graph and make it hard for the developer to discover. This is why PyTorch is more human-friendly and easy to adapt than other systems. Lastly, for Research speed, PyTorch allows the developer to generate new ideas quickly. Testing many different ways of cleaning and labelling the data is a must in a data-centric approach (Jakubuk et al., 2024). By using the PyTorch, it lets the developer see their results and be able to make changes to code instantly.

5.0 CONCLUSION

In conclusion, this paper explained why PyTorch is a leading technology for Data-Centric AI. The main part in this paper discussed that dynamic graph architecture is the thing that makes it faster and flexible for data processing. The main finding from the comparative analysis also shows that PyTorch is a better technology compared to legacy static systems as it is easier to fix and able to adapt to different data types. One limitation of PyTorch is that it may use more computer memory compared to other technology as it needs to rebuild the graph every time. Therefore, the updated technologies are recommended to be used in future research to ensure faster and ease of data processing.

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Date of Work	Search Keywords	Databases Used	AI Prompts Used	Task Description
21/4/2026	- Data centric AI system pytorch - Data centric AI system tensorflow	Scopus Web of Science Scopus AI	Data Centric AI system using PyTorch/ TensorFlow technology	- Search Scopus for papers related to data centric AI system that focused on PyTorch/ TensorFlow. - Find minimum 7 papers to read. - Points out notes from the papers to be included in the research
23/4/2026	- Suitable title for research	Gemini	Suggest me title of research that is focus on data centric AI and focus on PyTorch technology	- Select the most suitable title for research - Recap the notes that have been draft - Start doing the academic writing from intro, and technology description and architecture
24/4/2026	None	None	None	- Continue on the main body parts, conclusion, abstract and keywords
26/4/2026	None	Citation generator	Copy and paste the link of papers to citation generator to generate APA format references	- Updating the references part according to APA format - Proofread the paper and refine some of the parts. - Upload the pdf to Turnitin, and elearning for submission criteria.